

Q1. What is the KDD process in Data Mining? Explain with a suitable diagram.

Answer:

KDD stands for **Knowledge Discovery in Databases**.

In very simple words:

KDD is the process of taking a large amount of raw data and finding useful knowledge or information from it.

Think of it like this:

Raw Data → Clean Data → Useful Pattern → Knowledge

Steps of KDD

Remember these **6 steps**:

1. Data Selection

Choose the useful data from a large database.

2. Data Cleaning / Preprocessing

Remove errors, missing values, duplicate data, etc.

3. Data Transformation

Convert the data into a suitable format so it can be analyzed easily.

4. Data Mining

Apply techniques to find **patterns, relationships, or useful information** in the data.

5. Pattern Evaluation

Check which patterns are actually **useful and interesting**.

6. Knowledge Presentation

Show the final knowledge using **reports, graphs, charts, or other understandable forms**.

Easy Diagram

Database

↓

Data Selection



Data Cleaning



Data Transformation



Data Mining



Pattern Evaluation



Knowledge / Information

Easy Example

Suppose a supermarket has thousands of customer bills.

KDD can analyze those bills and discover:

Customers who buy bread often also buy butter.

This discovered relationship is **useful knowledge**.



Remember in One Line

Select → Clean → Transform → Mine → Evaluate → Present

S C T M E P

Think:

“Select Clean Things, Mine, Evaluate, Present.”



Exam Definition

KDD is the systematic process of discovering useful, valid, and understandable knowledge or patterns from large databases.

9/10/2026, 9:09:25 AM



Q2. Explain the following in OLAP:

- a) Roll Up Operation
- b) Drill Down Operation
- c) Slice Operation
- d) Dice Operation
- e) Pivot Operation

Answer:

First, understand **OLAP**.

OLAP = Online Analytical Processing

OLAP is used to **analyze large amounts of data from different angles**.

For example, a company has sales data for:

- Products
- Cities
- Months
- Years

OLAP helps us look at this data in different ways.

a) Roll Up Operation

Roll Up means going from detailed data to summarized data.

Example:

City → State → Country

Suppose we have sales:

- Ahmedabad = ₹10,000
- Surat = ₹15,000
- Vadodara = ₹8,000

Roll Up can combine these into:

Gujarat = ₹33,000

👉 **Remember:**

Roll Up = Detail → Summary

Think of **rolling up** data to a higher level.

b) Drill Down Operation

Drill Down is the opposite of Roll Up.

It means going from **summary data to more detailed data**.

Example:

Country → State → City

If we have:

India = ₹10,00,000

Drill Down can show:

- Gujarat = ₹3,00,000
- Maharashtra = ₹4,00,000
- Rajasthan = ₹3,00,000

Then we can drill further:

Gujarat → Ahmedabad, Surat, Vadodara

👉 **Remember:**

Drill Down = Summary → Detail

c) Slice Operation

Slice means selecting one particular value from one dimension.

Example:

Suppose sales data has:

Product × City × Year

If we select only:

Year = 2025

we get sales data only for 2025.

👉 **Remember:**

Slice = Select ONE

d) Dice Operation

Dice means selecting multiple values from multiple dimensions.

Example:

We select:

- Products = Laptop, Mobile
- Cities = Ahmedabad, Surat
- Years = 2024, 2025

Now we get a smaller portion of the original data.

👉 **Remember:**

Dice = Select MULTIPLE

e) Pivot Operation

Pivot means changing the way we view the data.

It **rotates the data dimensions** to get a different view.

Example:

Initially:

Products → Rows

Cities → Columns

After Pivot:

Cities → Rows

Products → Columns

The data is the same, but its **view/arrangement changes**.

👉 **Remember:**

Pivot = Change the View

🧠 **SUPER-EASY MEMORY TRICK**

Operation	Easy Meaning	Remember
Roll Up	Detail → Summary	⬆️ Summarize
Drill Down	Summary → Detail	⬇️ Details
Slice	Select one	ONE
Dice	Select many	MULTIPLE
Pivot	Change arrangement	VIEW

★ One-line memory

Roll Up = Summary, Drill Down = Detail, Slice = One, Dice = Many, Pivot = Change View.

9/10/2026, 9:09:37 AM

Q3. Define Data Warehouse Architecture with a suitable diagram.

Answer:

A **Data Warehouse** is a large storage system that collects data from different sources and stores it in one place for **analysis and decision-making**.

Data Warehouse Architecture means the **structure showing how data moves from different sources to the Data Warehouse and finally to the user**.

Basic Architecture

Data Sources

↓

ETL (Extract → Transform → Load)

↓

Data Warehouse

↓

OLAP / Analysis

↓

Users / Reports

1. Data Sources

These are the places from where data comes.

Examples:

- Databases
- Sales systems
- Excel files
- Websites
- Business applications

2. ETL

ETL = Extract, Transform, Load

- **Extract:** Take data from different sources.

- **Transform:** Clean and convert the data into the required format.
- **Load:** Put the prepared data into the Data Warehouse.

3. Data Warehouse

This is the **central storage** where large amounts of organized business data are stored.

4. OLAP

OLAP is used to analyze the stored data from different views.

For example:

- Sales by year
- Sales by city
- Sales by product

5. Users

Managers, analysts, and other users use the processed data to create:

- Reports
- Charts
- Graphs
- Business decisions

Easy Example

Imagine a company has sales data in **Delhi, Mumbai, and Ahmedabad**.

The data comes from different systems → **ETL cleans it** → everything is stored in the **Data Warehouse** → **OLAP analyzes it** → manager gets a **sales report**.

★ Remember in One Line

Sources → ETL → Data Warehouse → OLAP → Users

This is the basic flow you should remember for the exam.

9/10/2026, 9:09:50 AM

Q4. What is Data Cleaning? Discuss the methods adopted for outlier detection.

Answer:

Data Cleaning means **finding and fixing incorrect, incomplete, duplicate, or unwanted data** before using it for analysis.

In simple words:

Data Cleaning = Making dirty data clean and usable.

Why is Data Cleaning needed?

Real-world data often contains:

- Missing values
- Wrong values
- Duplicate records
- Outliers
- Inconsistent data

If we use dirty data, our final analysis can be **wrong**.

What is an Outlier?

An **outlier** is a data value that is **very different from the other values**.

Example:

10, 12, 11, 13, 12, 100

Here, **100** is an outlier because it is much larger than the other values.

Methods for Outlier Detection

1. Statistical Method

We use statistical measures such as:

- **Mean**

- **Standard deviation**
- **Z-score**

If a value is very far from the mean, it may be an outlier.

Example:

Most students score around 60–80, but one student scores 5.
That value may be an outlier.

2. Distance-Based Method

We calculate the **distance between data points**.

If one point is very far away from the other points, it can be considered an outlier.

Easy idea:

Far away from others = Possible outlier

3. Density-Based Method

We check how closely data points are grouped.

- Points in a **dense group** → normal
- Points in a **low-density area** → possible outlier

For example, if 100 customers have similar buying behavior but one customer behaves completely differently, that customer may be an outlier.

4. Clustering Method

Data is divided into groups called **clusters**.

If a data point does not properly belong to any cluster, it may be an outlier.

Easy Memory Trick

Remember the four methods as:

S-D-D-C

- **S** → Statistical
- **D** → Distance-based
- **D** → Density-based
- **C** → Clustering

★ Exam Definition

Data Cleaning is the process of detecting and correcting errors, missing values, duplicate data, inconsistent data, and outliers in a dataset to improve data quality.

An outlier is a data value that is significantly different from the other values in the dataset.

9/10/2026, 9:10:04 AM

Q5. What is Sampling in Data Reduction? Explain different types of Probability and Non-Probability Sampling methods.

Answer:

What is Sampling?

Sampling means selecting a **small part of a large dataset** and using it to understand the whole dataset.

In simple words:

Instead of studying everyone, we study a smaller group that represents everyone.

Easy Example

Suppose a college has **10,000 students** and we want to know their opinion about online classes.

It is difficult to ask all 10,000 students.

So, we select **500 students** and ask them.

Here:

- **10,000 students = Population**
- **500 selected students = Sample**

This helps reduce **time, cost, and processing effort**.

Types of Sampling

There are mainly **two types**:

- 1. Probability Sampling**
- 2. Non-Probability Sampling**

1. Probability Sampling

In probability sampling, **every member has a known chance of being selected**.

a) Simple Random Sampling

Members are selected **completely randomly**.

Example:

From 1,000 students, randomly select 100 students.

👉 **Remember:** Random = everyone gets a chance.

b) Systematic Sampling

Select members at a **fixed interval**.

Example:

Select every **10th student** from a list:

10 → 20 → 30 → 40 → 50...

👉 **Remember:** Systematic = **fixed gap**.

c) Stratified Sampling

First divide the population into **groups (strata)**, then randomly select members from each group.

Example:

College students are divided into:

- First year
- Second year
- Third year

Then students are selected from each group.

👉 **Remember:** Stratified = **divide into groups, then select**.

d) Cluster Sampling

Divide the population into **clusters/groups**, then select some complete clusters.

Example:

A city has 100 schools.

Select 10 schools and survey **all students in those selected schools**.

👉 **Remember:** Cluster = **select whole groups**.

2. Non-Probability Sampling

In non-probability sampling, **everyone does not have a known or equal chance of being selected**.

a) Convenience Sampling

Select people who are **easiest to reach**.

Example:

Ask students who are sitting near you.

👉 **Remember:** Convenience = **easy to reach**.

b) Judgment / Purposive Sampling

The researcher selects people based on **their own judgment** because they are suitable for the study.

Example:

To study professional programmers, select experienced programmers.

👉 **Remember:** Judgment = **researcher chooses**.

c) Quota Sampling

Divide people into groups and select a **fixed number (quota)** from each group.

Example:

Select:

- 50 males
- 50 females

👉 **Remember:** Quota = **fixed number**.

d) Snowball Sampling

One participant recommends or brings **another participant**, and the process continues.

Example:

You interview one expert → that expert introduces another expert → that expert introduces another.

👉 **Remember:** Snowball = **one person leads to another.**

🧠 SUPER-EASY MEMORY TABLE

Type	Meaning
Simple Random	Select randomly
Systematic	Select at fixed intervals
Stratified	Divide into groups, select from each
Cluster	Select complete groups
Convenience	Select easiest people
Judgment	Researcher decides
Quota	Select a fixed number
Snowball	One participant finds another

★ One-Line Memory Trick

Probability:

Random → Gap → Groups → Clusters

Non-Probability:

Easy → Judgment → Fixed Number → Chain

★ Exam Definition

Sampling is the process of selecting a smaller representative subset from a large population or dataset for analysis, thereby reducing time, cost, and processing requirements.

9/10/2026, 9:10:25 AM

Q6. What is the process of Attribute Subset Selection? Explain the following methods of attribute subset selection:

- a) Stepwise Forward Selection**
- b) Stepwise Backward Elimination**
- c) Combination of Forward Selection and Backward Elimination**

Answer:

What is Attribute Subset Selection?

Attribute Subset Selection means **selecting only the important attributes (features) from a dataset and removing unnecessary ones.**

In simple words:

Keep useful columns, remove useless columns.

Easy Example

Suppose we want to predict a student's exam marks.

Our dataset has:

- Name
- Age
- Attendance
- Study Hours
- Favourite Colour
- Phone Model
- Previous Marks

For predicting marks, **Attendance, Study Hours, and Previous Marks** may be useful.

Name, Favourite Colour, and Phone Model may not be useful.

So we keep the useful attributes and remove the unnecessary ones.

a) Stepwise Forward Selection

We **start with no attributes**.

Then we add the **most useful attribute one by one**.

Example:

Start:

Nothing

↓ Add Study Hours

Study Hours

↓ Add Attendance

Study Hours + Attendance

↓ Add Previous Marks

Study Hours + Attendance + Previous Marks

👉 **Remember: Forward = Start Empty → Keep Adding**

b) Stepwise Backward Elimination

We **start with all attributes**.

Then we remove the **least useful attribute one by one**.

Example:

Start:

Age + Attendance + Study Hours + Favourite Colour + Phone Model + Previous Marks

↓ Remove Phone Model

↓ Remove Favourite Colour

↓ Remove Age

Final:

Attendance + Study Hours + Previous Marks

👉 **Remember: Backward = Start Full → Keep Removing**

c) Combination of Forward Selection and Backward Elimination

This method uses **both methods together**.

First, we **add useful attributes**.

Then, we **check the selected attributes and remove any attribute that has become unnecessary**.

Example:

Start with nothing.

Add Study Hours

↓

Add Attendance

↓

Add Previous Marks

↓

Check again

↓

If Attendance is no longer useful → **Remove Attendance**

Final:

Study Hours + Previous Marks

👉 **Remember: Add + Remove = Combination**

🧠 SUPER-EASY MEMORY

Method	Starts With	Action
Forward Selection	Nothing	Add useful attributes
Backward Elimination	Everything	Remove useless attributes
Combination	Nothing/selected set	Add + Remove

★ One-Line Memory Trick

Forward = Add → Backward = Remove → Combination = Add + Remove

★ Exam Definition

Attribute Subset Selection is the process of selecting the most relevant attributes from a dataset and removing irrelevant or redundant attributes to reduce the size and complexity of data.

9/10/2026, 9:10:39 AM

Q7. What is Market Basket Analysis? Explain the two measures of rule interestingness: Support and Confidence.

Answer:

What is Market Basket Analysis?

Market Basket Analysis is a technique used to find **which items are commonly purchased together** by customers.

In simple words:

It finds the relationship between items that people buy together.

Easy Example

Suppose a supermarket checks many customer bills and finds:

Bread + Butter are frequently bought together.

So the store may place **bread and butter near each other** or offer them as a combination.

This type of analysis is called **Market Basket Analysis**.

Two Measures of Rule Interestingness

To check whether an association rule is useful, we mainly use:

1. **Support**
2. **Confidence**

1. Support

Support tells us how frequently the items occur together in the entire dataset.

Formula:

Support(A → B) =

Number of transactions containing A and B / Total number of transactions

Example

Suppose there are **100 transactions**.

In **20 transactions**, customers bought both:

Bread + Butter

Then:

$$\text{Support} = 20 / 100 = 20\%$$

So, Bread and Butter occur together in **20% of all transactions**.

👉 **Remember:**

Support = How often they occur together.

2. Confidence

Confidence tells us how often B is purchased when A is purchased.

Formula:

$$\text{Confidence}(A \rightarrow B) =$$

Number of transactions containing A and B / Number of transactions containing A

Example

Suppose:

- 40 customers bought Bread
- 20 of those customers also bought Butter

Then:

$$\text{Confidence} = 20 / 40 = 50\%$$

This means:

50% of customers who bought Bread also bought Butter.

👉 **Remember:**

Confidence = If A is bought, how likely is B to be bought?

SUPER-EASY MEMORY

Measure	Meaning
Support	How often A and B occur together
Confidence	How often B happens when A happens

★ **One-Line Memory Trick**

Support = Together

Confidence = If A, then B

★ **Exam Definition**

Market Basket Analysis is a data mining technique used to discover relationships and purchasing patterns among items frequently bought together.

Support measures how frequently an itemset occurs in the complete dataset, while confidence measures how frequently B is purchased when A is purchased.

9/10/2026, 9:10:49 AM

Q8. State the Apriori Property. Generate large itemsets and association rules using the Apriori algorithm.

Answer:

The **Apriori Property** is a rule used in **Association Rule Mining**.

If an itemset is frequent, then all of its subsets must also be frequent.

In simple words:

If a group of items is bought frequently, then every smaller group inside it must also be bought frequently.

Easy Example

Suppose:

{Bread, Milk, Butter} is a frequent itemset.

Then these must also be frequent:

- {Bread, Milk}
- {Bread, Butter}
- {Milk, Butter}

If **{Bread, Milk}** is not frequent, then **{Bread, Milk, Butter}** cannot be frequent.

Remember

Frequent BIG set → Frequent SMALL sets

And the opposite is also useful:

If a small set is NOT frequent → its bigger set cannot be frequent.

This property helps Apriori **remove unnecessary combinations**, so it does not have to check every possible item combination.

Apriori Algorithm — Basic Steps

Suppose we have items:

A, B, C, D

Step 1: Find 1-itemsets

Check each individual item:

{A}, {B}, {C}, {D}

Remove the items that do not satisfy minimum support.

The remaining items are called **frequent 1-itemsets (L1)**.

Step 2: Generate 2-itemsets

Combine frequent 1-itemsets:

{A,B}, {A,C}, {A,D}, {B,C}...

Calculate their support.

Keep only the ones satisfying minimum support.

These are **frequent 2-itemsets (L2)**.

Step 3: Generate 3-itemsets

Use frequent 2-itemsets to create 3-itemsets.

Again, calculate support and remove combinations that do not satisfy minimum support.

Step 4: Continue

Keep generating larger itemsets until **no more frequent itemsets can be found**.

Generating Association Rules

After finding frequent itemsets, we generate rules such as:

A → B

This means:

*If a customer buys **A**, they may also buy **B**.*

Then calculate:

Support → How frequently A and B occur together.

Confidence → How often B occurs when A occurs.

Keep the rule if it satisfies the **minimum support and minimum confidence** requirements.

Easy Memory Trick

Remember Apriori as:

Find → Combine → Check → Remove → Repeat → Make Rules

Exam Definition

Apriori Property states that all non-empty subsets of a frequent itemset must also be frequent. The Apriori algorithm uses this property to efficiently find frequent itemsets and generate association rules from large datasets.

9/10/2026, 9:11:10 AM

Q9. Write short notes on:

- a) Single Dimensional Association Rule
- b) Decision Tree Induction

Answer:

a) Single Dimensional Association Rule

A **Single Dimensional Association Rule** is an association rule that involves **only one type of attribute or dimension**.

In simple words:

It finds relationships between items using only one dimension, usually the items purchased.

Example

Suppose a supermarket studies only **purchased items**.

It finds:

Bread → Butter

This means:

Customers who buy bread often also buy butter.

Here, we are considering only **one dimension: purchased items**.

Remember

Single Dimension = One type of information

b) Decision Tree Induction

Decision Tree Induction is a technique used to build a **decision tree from data** for classification or prediction.

A **Decision Tree** looks like a tree:

- **Root** → Starting question

- **Branches** → Possible answers
- **Leaf** → Final decision/class

Easy Example

Suppose we want to decide whether a student will **Pass or Fail**.

The tree may ask:

Attendance > 75%?

↓ Yes → **Study Hours > 2?**

→ Yes → **Pass**

→ No → **Fail**

↓ No → **Fail**

The computer learns these questions from existing data and creates the tree.

Basic Process

Training Data

↓

Select Best Attribute

↓

Create Decision Node

↓

Split the Data

↓

Repeat

↓

Final Decision Tree



Remember

Decision Tree = Ask Questions → Follow Branches → Get Decision

★ Exam Definitions

Single Dimensional Association Rule:

An association rule that involves only one dimension or type of attribute, such as items purchased.

Decision Tree Induction:

Decision Tree Induction is the process of constructing a decision tree from training data for classification or prediction.

9/10/2026, 9:11:31 AM

Q10. What is Bayesian Classification? Explain Naive Bayes Theorem with a suitable example.

Answer:

What is Bayesian Classification?

Bayesian Classification is a classification method based on **Bayes' Theorem**.

It is used to predict **which class a data item belongs to**.

In simple words:

Bayesian Classification uses probability to make a decision.

Easy Example

Suppose we want to predict whether an email is:

- **Spam**
- **Not Spam**

The system looks at things such as:

- Contains the word "offer"
- Contains the word "free"
- Contains suspicious links

Then it calculates the probability of the email being **Spam** and **Not Spam**.

Whichever probability is higher becomes the prediction.

What is Naive Bayes?

Naive Bayes is a simple and popular Bayesian classification algorithm.

It is called "**Naive**" because it assumes that the different features are **independent of each other**.

For example, it assumes:

*The presence of the word "**free**" is independent of the presence of the word "**offer**".*

This assumption makes the calculation much easier.

Bayes' Theorem

The basic formula is:

$$P(A | B) = [P(B | A) \times P(A)] / P(B)$$

Where:

- $P(A | B)$ = Probability of A when B is known
- $P(B | A)$ = Probability of B when A is known
- $P(A)$ = Probability of A
- $P(B)$ = Probability of B

Easy way to understand it

Suppose:

A = Email is Spam

B = Email contains "Free"

Then:

P(Spam | Free)

means:

What is the probability that the email is Spam if it contains "Free"?

How Naive Bayes Works

Remember these **4 steps**:

1. Collect the data

↓

2. Calculate probabilities

↓

3. Compare probabilities for each class



4. Choose the class with the highest probability

Example

Suppose an email contains:

"Free offer"

Naive Bayes calculates:

Probability of Spam = 80%

Probability of Not Spam = 20%

Since **80% > 20%**, the email is classified as:

SPAM

Easy Memory Trick

Bayesian = Probability-based classification

Naive Bayes = Probability + Independent Features

★ Exam Definition

Bayesian Classification is a classification technique based on Bayes' Theorem that uses probability to predict the class of a data object.

Naive Bayes assumes that all features are independent of each other and calculates the probability of each class to make the final prediction.

9/10/2026, 9:11:42 AM

Q11. Discuss Linear and Non-Linear Regression with suitable examples.

Answer:

What is Regression?

Regression is a data mining technique used to **predict a numerical value** based on existing data.

In simple words:

Regression = Use existing data to predict a number.

Easy Example

Suppose we want to predict a **house price** based on its size.

- 500 sq. ft. → ₹20 lakh
- 1000 sq. ft. → ₹40 lakh
- 1500 sq. ft. → ₹60 lakh

Regression can help us predict the price of a **1200 sq. ft. house**.

1. Linear Regression

Linear Regression is used when the relationship between variables is approximately a **straight line**.

Simple idea:

X increases → Y also increases/decreases in a straight-line pattern.

Example:

House Size → House Price

As house size increases, price increases approximately in a straight-line relationship.

Formula:

$$y = a + bx$$

Where:

- **y** = predicted value
- **x** = input value
- **a** = intercept
- **b** = slope

Example

If:

$$y = 10 + 2x$$

and **x** = 5:

$$y = 10 + (2 \times 5)$$

$$y = 20$$

So, predicted value = **20**.

👉 **Remember:**

Linear Regression = Straight Line

2. Non-Linear Regression

Non-Linear Regression is used when the relationship between variables **does not follow a straight line**.

The relationship may form a **curve**.

Example

Consider:

Age of a person → Height

Height increases rapidly during childhood, then increases slowly, and finally stops increasing.

This relationship is not a simple straight line, so a **non-linear model** may be more suitable.

Another example:

Advertising expenditure → Sales

Increasing advertising may increase sales, but after a certain point, the increase may become smaller.

👉 **Remember:**

Non-Linear Regression = Curved relationship

Difference Between Linear and Non-Linear Regression

Linear Regression

Relationship is approximately a **straight line**

Simple relationship

Example: Size → Price

Formula often uses $y = a + bx$

Non-Linear Regression

Relationship is **curved/non-linear**

More complex relationship

Example: Growth → Age

Uses a non-linear mathematical function

🧠 SUPER-EASY MEMORY

Linear = Line

Non-Linear = Curve

★ Exam Definition

Linear Regression predicts a numerical value when the relationship between variables can be represented approximately by a straight line.

Non-Linear Regression predicts a numerical value when the relationship between variables cannot be represented by a straight line and follows a curved or more complex pattern.

9/10/2026, 9:11:55 AM

Q12. What is Association Rule Mining? Explain it with an example.

Answer:

What is Association Rule Mining?

Association Rule Mining is a data mining technique used to **find relationships or connections between different items in a large dataset.**

In simple words:

It finds what things are commonly found or bought together.

Easy Example

Suppose a supermarket studies customer bills.

It finds that many customers who buy:

Bread → also buy Butter

So we can write an association rule:

Bread → Butter

This means:

*Customers who buy **Bread** are likely to also buy **Butter**.*

Parts of an Association Rule

An association rule usually has this form:

A → B

Where:

- **A = Antecedent** → the item already present
- **B = Consequent** → the item associated with it

Example:

Milk → Bread

- **Milk** = Antecedent

- **Bread** = Consequent

Why is it useful?

Association Rule Mining helps businesses:

- Understand customer buying habits
- Recommend products
- Place related products together
- Create product offers and combinations

🧠 Easy Example to Remember

Imagine 100 supermarket bills.

If many bills contain:

Milk + Bread + Butter

the system can discover rules such as:

Milk → Bread

or

Bread → Butter

These are called **association rules**.

★ Exam Definition

Association Rule Mining is a data mining technique used to discover interesting relationships and patterns among items in a large dataset.

🧠 One-Line Memory

Association Rule Mining = Finding what things occur together.

9/10/2026, 9:12:05 AM

Q13. Explain Data Warehouse Architecture with a suitable diagram.

Answer:

A **Data Warehouse Architecture** shows how data is collected from different sources, processed, stored, and finally used for analysis.

Basic Structure

Data Sources

↓

ETL Process

↓

Data Warehouse

↓

OLAP / Analysis Tools

↓

Users

1. Data Sources

Data comes from different places, such as:

- Databases
- Excel files
- Sales systems
- Websites
- Business applications

2. ETL Process

ETL = Extract → Transform → Load

- **Extract:** Collect data from different sources.
- **Transform:** Clean and convert the data into a proper format.
- **Load:** Store the prepared data in the Data Warehouse.

3. Data Warehouse

It is the **central storage** where large amounts of organized data are stored.

4. OLAP / Analysis

OLAP tools analyze the stored data.

For example:

- Sales by year
- Sales by city
- Sales by product

5. Users

Managers and analysts use the information to create:

- Reports
- Charts
- Graphs
- Business decisions

★ Exam Diagram

Different Data Sources



Extract



Transform



Load



OLAP / Analysis



Reports / Users

🧠 Easy Memory Trick

Source → ETL → Warehouse → OLAP → User

ETL = Extract → Transform → Load

★ Exam Definition

Data Warehouse Architecture is the structure that shows how data is collected from different sources, processed using ETL, stored in a central data warehouse, analyzed using OLAP, and presented to users.

9/10/2026, 9:12:29 AM

Q13. What is Decision Tree Induction?

Answer:

We already covered **Decision Tree Induction** in Q9, so this theory question is **completed**.

The remaining theory topics are also completed. ✅

🟢 THEORY COMPLETE

1. KDD Process ✅
2. OLAP Operations ✅
3. Data Warehouse Architecture ✅
4. Data Cleaning & Outlier Detection ✅
5. Sampling in Data Reduction ✅
6. Attribute Subset Selection ✅
7. Market Basket Analysis, Support & Confidence ✅
8. Apriori Property & Algorithm — theory part ✅
9. Single Dimensional Association Rule ✅
10. Decision Tree Induction ✅
11. Bayesian Classification / Naive Bayes ✅
12. Linear & Non-Linear Regression ✅
13. Association Rule Mining ✅

Now we should start the Numerical/Sum section.

9/10/2026, 9:13:50 AM

Q1. Perform data smoothing using equal-frequency (equi-depth) binning for the data:

4, 8, 15, 21, 24, 25, 28, 34

Perform smoothing using:

a) Bin Means

b) Bin Boundaries

9/10/2026, 9:14:18 AM

Q1. Perform data smoothing using equal-frequency (equi-depth) binning for the data: 4, 8, 15, 21, 24, 25, 28, 34.

Answer:

This is a **numerical question**, and before solving it, let's understand what it is asking.

What is Data Smoothing?

Data smoothing means reducing small irregularities or noise in data so that the data becomes easier to understand.

One common method is **Binning**.

What is Binning?

Binning means:

Divide the data into small groups called bins, then replace values in each bin with a representative value.

Here our data is:

4, 8, 15, 21, 24, 25, 28, 34

The question says to use **equal-frequency (equi-depth) binning**.

What does Equal-Frequency mean?

It means:

Each bin must contain the same number of data values.

We have **8 values** and the question asks for **3 bins**.

Since 8 cannot be divided equally into 3, we divide them as evenly as possible:

- **Bin 1:** 4, 8, 15
- **Bin 2:** 21, 24, 25
- **Bin 3:** 28, 34

Now we perform **data smoothing**.

a) Smoothing by Bin Means

Mean = Sum of values ÷ Number of values

Bin 1

Values:

4, 8, 15

Mean:

$$(4 + 8 + 15) / 3 = 27 / 3 = 9$$

So:

4, 8, 15 → 9, 9, 9

Bin 2

Values:

21, 24, 25

Mean:

$$(21 + 24 + 25) / 3 = 70 / 3 = 23.33$$

So:

21, 24, 25 → 23.33, 23.33, 23.33

Bin 3

Values:

28, 34

Mean:

$$(28 + 34) / 2 = 31$$

So:

28, 34 → 31, 31

Final Answer — Bin Means

Original:

4, 8, 15 | 21, 24, 25 | 28, 34

After smoothing:

9, 9, 9 | 23.33, 23.33, 23.33 | 31, 31

b) Smoothing by Bin Boundaries

For **bin boundary smoothing**, we replace each value with the **nearest boundary value**.

The boundaries are the **smallest and largest values in each bin**.

Bin 1

4, 8, 15

Boundaries = **4 and 15**

- 4 → 4
- 8 → 4 (closer to 4)
- 15 → 15

Result:

4, 4, 15

Bin 2

21, 24, 25

Boundaries = **21 and 25**

- 21 → 21
- 24 → 25 (closer to 25)
- 25 → 25

Result:

21, 25, 25

Bin 3

28, 34

Boundaries = **28 and 34**

- 28 → 28
- 34 → 34

Result:

28, 34

Final Answer — Bin Boundaries

4, 4, 15 | 21, 25, 25 | 28, 34

🧠 VERY EASY WAY TO REMEMBER

Binning question:

First divide → then smooth.

Bin Mean:

*Replace everything with the **average**.*

Bin Boundary:

*Replace everything with the **nearest edge value**.*

★ Exam Memory Trick

Mean = Average

Boundary = Nearest End

So:

Bin Mean → Average of the bin

Bin Boundary → Nearest boundary of the bin

9/10/2026, 9:14:41 AM

Q2. Explain Data Normalisation. Use the following methods to Normalise the data: 200, 300, 400, 600, 1000.

Answer:

This is a **numerical question**. We have to normalize the given data using **3 different methods**:

1. **Min-Max Normalisation**
2. **Z-score Normalisation**
3. **Decimal Scaling Normalisation**

First understand what **normalisation** means.

What is Data Normalisation?

Data Normalisation means converting values into a smaller and standard range so that they can be easily compared.

For example, our values are:

200, 300, 400, 600, 1000

These values have a large range. Normalisation converts them into smaller values.

a) Min-Max Normalisation for \$400

Given:

- **$x = 400$**
- **Minimum = 200**
- **Maximum = 1000**
- **new_min = 0**
- **new_max = 1**

Formula

New value = $[(x - \text{min}) / (\text{max} - \text{min})] \times (\text{new_max} - \text{new_min}) + \text{new_min}$

Put the values:

$$\begin{aligned} &= [(400 - 200) / (1000 - 200)] \times (1 - 0) + 0 \\ &= (200 / 800) \times 1 \\ &= 0.25 \end{aligned}$$

Final Answer:

\$400 → 0.25

👉 **Remember:**

$$\text{Min-Max} = \text{Current} - \text{Minimum} \div \text{Maximum} - \text{Minimum}$$

b) Z-score Normalisation for \$300

For Z-score, we use:

$$\mathbf{Z = (x - \text{Mean}) / Standard Deviation}$$

```
genui{"learning_viz":{"type_id":"STANDARD_SCORE_Z","initial_values":  
{"x":300,"mu":500,"sigma":303.315}}}
```

Step 1: Find Mean

Data:

200, 300, 400, 600, 1000

Mean:

$$\begin{aligned} &(\mathbf{200 + 300 + 400 + 600 + 1000}) / 5 \\ &= \mathbf{2500 / 5} \\ &= \mathbf{500} \end{aligned}$$

So:

$$\mathbf{Mean = 500}$$

Step 2: Find Standard Deviation

Using the population standard deviation:

$$\sigma \approx \mathbf{303.32}$$

Step 3: Find Z-score

For \$300:

$$Z = (300 - 500) / 303.32$$

$$Z = -200 / 303.32$$

$$Z \approx -0.66$$

Final Answer:

$$\text{\$300} \rightarrow -0.66$$

👉 The negative sign means **\\$300 is below the mean.**

c) Decimal Scaling Normalisation for \$600

Formula

$$v' = v / 10^k$$

Where **k** is the smallest integer such that the largest normalized value is less than 1.

Our largest value is:

1000

We need:

$$1000 / 10^3 = 0.1$$

Therefore:

$$k = 3$$

Now normalize **600**:

$$600 / 10^3$$

$$= 600 / 1000$$

$$= 0.6$$

Final Answer:

$$\text{\$600} \rightarrow 0.6$$

★ FINAL ANSWERS

Method	Asked Value	Answer
Min-Max	\$400	0.25
Z-score	\$300	-0.66
Decimal Scaling	\$600	0.6

🧠 ONE-TIME MEMORY TRICK

Min-Max:

Subtract minimum → Divide by range

Z-score:

Subtract mean → Divide by standard deviation

Decimal Scaling:

Move decimal by dividing by 10, 100, 1000...

So remember:

MIN-MAX = MIN

Z-SCORE = MEAN

DECIMAL = 10^k

9/10/2026, 9:14:58 AM

Q3. State the Apriori Property. Generate large itemsets and association rules using the Apriori algorithm.

Answer:

This is the **last numerical question**. It is a little longer, so first understand the basic idea.

What is Apriori?

Apriori is an algorithm used to find frequently purchased groups of items in a database.

For example, in a supermarket:

Bread + Milk may be purchased together many times.

Apriori helps us find such **frequent itemsets** and then create **association rules**.

Given Data

Transaction	Items
T1	Bread, Cheese, Egg, Juice
T2	Bread, Cheese, Juice
T3	Bread, Milk, Yogurt
T4	Bread, Juice, Milk
T5	Cheese, Juice, Milk

Given:

Minimum Support = 50%

Minimum Confidence = 75%

Since there are **5 transactions**:

50% of 5 = 2.5

Therefore, an itemset must occur in **at least 3 transactions** to be considered frequent.

👉 Important:

For this question:

Minimum support count = 3

Step 1: Find 1-Itemsets

Count how many transactions contain each item.

Bread

T1, T2, T3, T4 = 4

Support:

4/5 = 80% ✓

Cheese

T1, T2, T5 = 3

Support:

3/5 = 60% ✓

Egg

T1 = 1

Support:

1/5 = 20% ✗

Juice

T1, T2, T4, T5 = 4

Support:

4/5 = 80% ✓

Milk

T3, T4, T5 = 3

Support:

$3/5 = 60\%$ ✓

Yogurt

$T3 = 1$

Support:

$1/5 = 20\%$ ✗

So we remove **Egg** and **Yogurt**.

Frequent 1-itemsets (L1)

{Bread}, {Cheese}, {Juice}, {Milk}

Step 2: Find 2-Itemsets

Now combine the frequent items.

Possible pairs:

- Bread + Cheese
- Bread + Juice
- Bread + Milk
- Cheese + Juice
- Cheese + Milk
- Juice + Milk

Now count them.

Bread + Cheese

$T1, T2 = 2$

Support = $2/5 = 40\%$ ✗

Remove.

Bread + Juice

T1, T2, T4 = 3

Support = $3/5 = 60\%$ ✓

Bread + Milk

T3, T4 = 2

Support = 40% ✗

Cheese + Juice

T1, T2, T5 = 3

Support = 60% ✓

Cheese + Milk

T5 = 1

Support = 20% ✗

Juice + Milk

T4, T5 = 2

Support = 40% ✗

Frequent 2-itemsets (L2)

{Bread, Juice}

{Cheese, Juice}

Step 3: Find 3-Itemsets

Now try to make a 3-itemset.

The only possible combination from our frequent pairs is:

Bread + Cheese + Juice

But according to Apriori Property:

Bread + Cheese is NOT frequent.

Therefore:

Bread + Cheese + Juice cannot be frequent.

So there are **no frequent 3-itemsets**.

Step 4: Generate Association Rules

Our frequent itemsets are:

{Bread, Juice}

{Cheese, Juice}

Now generate rules.

Rule 1:

Bread → Juice

Support:

3/5 = 60%

Confidence:

Support(Bread + Juice) / Support(Bread)

= 3/4

= 75% ✓

So:

Bread → Juice is a valid rule.

Rule 2:

Juice → Bread

Confidence:

3/4 = 75% ✓

So:

Juice → Bread is also valid.

Rule 3:

Cheese → Juice

Confidence:

3/3 = 100% 

So:

Cheese → Juice is valid.

Rule 4:

Juice → Cheese

Confidence:

3/4 = 75% 

So:

Juice → Cheese is valid.

★ Final Association Rules

Rule	Support	Confidence	Result
Bread → Juice	60%	75%	
Juice → Bread	60%	75%	
Cheese → Juice	60%	100%	
Juice → Cheese	60%	75%	

How to Remember Apriori in Exam

Always follow this order:

1. Count 1-itemsets

↓

2. Remove items below minimum support

↓

3. Make 2-itemsets

↓

4. Remove weak pairs

↓

5. Make 3-itemsets

↓

6. Stop when no more frequent sets are possible

↓

7. Generate rules

↓

8. Check confidence

★ **One-line memory**

COUNT → REMOVE → COMBINE → CHECK → MAKE RULES

And remember the most important Apriori rule:

If a smaller itemset is not frequent, its larger itemset cannot be frequent.

9/10/2026, 9:16:26 AM